

TANAY DESAI

(667) 289-3387 tanaydesai2004@gmail.com [linkedin.com/in/tanaydesai1510/](https://www.linkedin.com/in/tanaydesai1510/) github.com/TanayDesai-1510 [Portfolio](#)

EDUCATION

Rutgers University–New Brunswick

Sep 2023 – present

Bachelor of Science in Computer Science

Shri Bhagubhai Mafatlal Polytechnic

Dec 2020 – Jun 2023

Diploma in Information Technology

RELEVANT COURSEWORK

- Data Structures
- Algorithms Analysis
- Artificial Intelligence
- Operating Systems
- Software Engineering
- Database Management
- Computer Network
- Computer Architecture

EXPERIENCE

Rutgers OIT-EAS - Software Engineer

Feb 2024 – present

React, Redux, Typescript, Javascript

New Brunswick, NJ

- Collaborating with cross-functional developers to modernize and optimize Rutgers University’s central web portal, an enterprise-scale platform supporting academic and student services, relied upon daily by over **70,000** students, faculty, and staff.
- Developing custom, reusable, and responsive React.js components, significantly reducing overall development time by **30%** and enhancing overall code maintainability, scalability, and cross-project performance across multiple applications.
- Resolving critical software issues, improving overall system performance by **15%** and providing a smoother user experience.
- Engineered dark mode features, increasing user engagement by **20%** while improving visual appeal and overall experience.

Skillsvista - Front-End Engineering Intern

Jul 2022 – Aug 2022

Vue, Vuex, Javascript

Dublin, Ireland

- Designed and developed **3+** highly responsive and reusable web components using Vue.js, significantly improving user experience and overall performance for over **2,000+** monthly users, while ensuring seamless functionality across diverse devices and platforms.
- Leveraged **Vuex** for robust and scalable state management, improving data flow, system stability, and overall application performance by ensuring consistent state synchronization across components and reducing potential errors.
- Identified, diagnosed, and efficiently fixed **5+** critical software bugs across multiple modules, resulting in a **15%** reduction in support tickets by significantly improving overall system stability, minimizing errors, and greatly enhancing the overall end-user experience.

PROJECTS

STAMP | *React.js, Python, Flask, Chakra-UI*

Sep 2023

- Spearheaded the end-to-end design and development of a fully responsive and dynamic user interface using **React.js** and **Chakra UI**, significantly reducing overall user friction, streamlining navigation, and boosting usability scores by **25%** through an intuitive, accessible, and seamless digital experience across multiple devices.
- Integrated RESTful Flask APIs to facilitate seamless and secure communication between the front-end and back-end systems, achieving a **95%** API response success rate while ensuring efficient data exchange and overall system reliability.

RU-ScarletSelector | *React.js, Python, Flask, Material-UI*

Mar 2024

- Built a course recommendation system at HackRU in under 24 hours, helping Rutgers students pick optimal courses; designed a React.js and Material-UI interface that improved onboarding by **30%** and developed Flask APIs with sub-200ms response times.

EPOCH-OS | *Rust, TypeScript, RAG, Docker, React, LLMs (Ollama/OpenAI), Tauri, ChromaDB, CI/CD*

Aug 2025

- Directed the frontend architecture and implementation using React, TypeScript, and Tauri to deliver a seamless, high-performance, and scalable cross-platform user experience, improving load times by **20%** and reducing UI-related defects by **15%**.
- Led the development of an intelligent operating environment that integrated AI agents and advanced Voice AI capabilities into digital workflows and productivity tools, automating repetitive tasks and improving overall efficiency by nearly **30%**.
- Built a Retrieval-Augmented Generation (RAG) pipeline leveraging ChromaDB and large language models (LLMs) to deliver highly domain-specific and context-aware AI responses, increasing query accuracy by nearly **35%** and reducing response latency by **20%**.

TECHNICAL SKILLS

Programming Languages: Java, Python, C/C++, JavaScript, TypeScript, SQL, HTML/CSS, R, C#, Rust

Libraries and Frameworks: React.js, Node.js, Next.js, Vue.js, Express.js, Flask, .NET, Tauri

Databases: MongoDB, MySQL, Oracle, PostgreSQL, SQL, SQLite

Tools & Platforms: Git, Github, Firebase, Postman, VScode, Docker, ChromaDB

ACHIEVEMENTS

- Orientation Leader for **6000+** students; awarded **Best Orientation Leader of the Session** for leadership excellence.
- President of PolyHacks, a flagship hackathon with **200+** participants; managed planning, sponsorships, and team coordination.
- Secured **Second Place** in a Competitive Programming Competition, showcasing strong problem-solving and teamwork skills.
- Elected **Student Representative** for Final Year Diploma; advocated student interests and improved faculty communication.